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output: pdf\_document: toc: no toc\_depth: '3' number\_sections: yes latex\_engine: xelatex  
html\_document: toc: yes toc\_depth: 3 toc\_float: yes number\_sections: yes highlight: tango theme:  
default fig\_caption: yes df\_print: tibble urlcolor: blue —

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.4.4      v tibble    3.2.1
## v lubridate  1.9.3      v tidyr     1.3.0
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

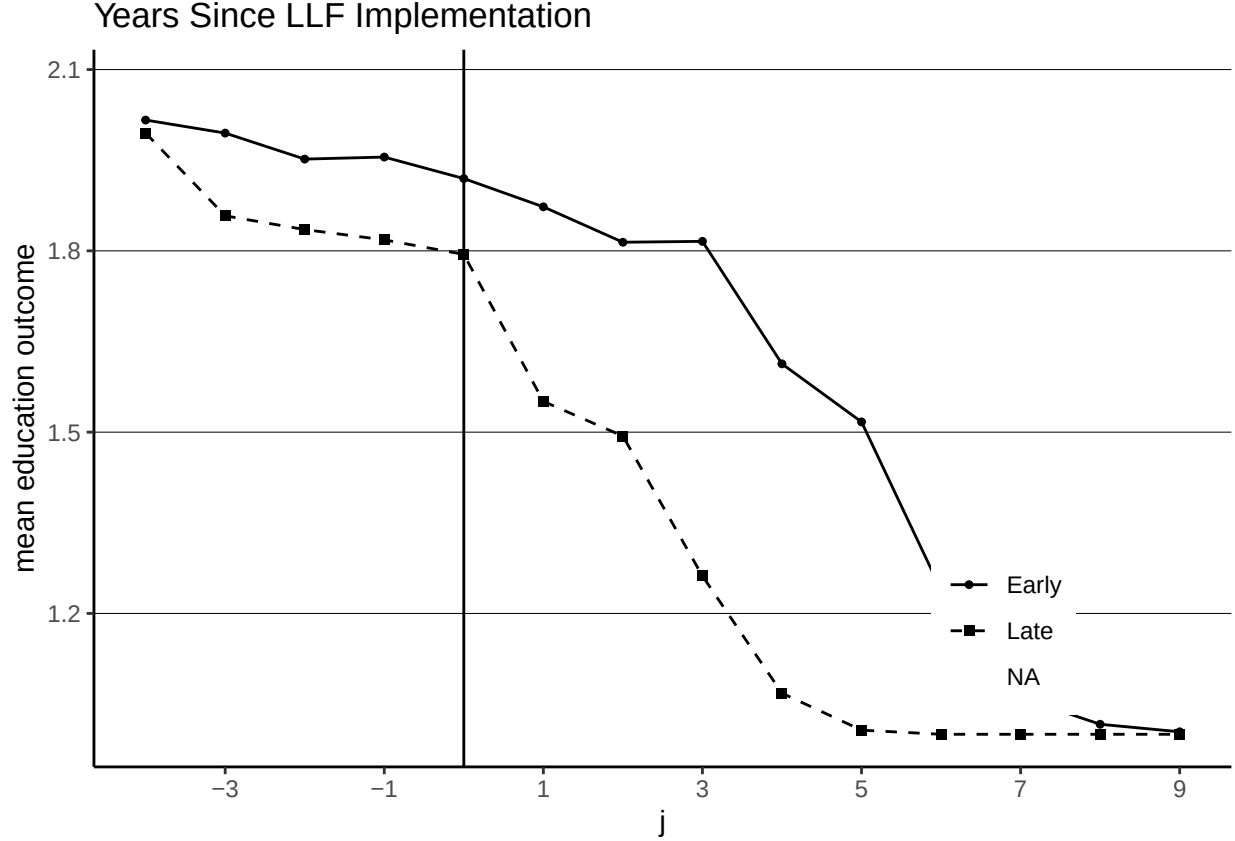
## [1] "/Users/kongying/Desktop/U6614-R/final project"
```

# 1 Empirical Strategy

## 1.1 Examine underlying parallel trend for DID model

```
## Warning: The `size` argument of `element_line()` is deprecated as of ggplot2 3.4.0.
## i Please use the `linewidth` argument instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.

## Warning: Removed 21 rows containing missing values (`geom_point()`).
```



The pre-treatment trends between “Early” and “Late” are not parallel before  $j=-1$ , which suggests that the assumption of parallel trends may not be satisfied.

**Parallel Trends Assumption:** In the absence of the LLF policy, the educational attainment trends for individuals in both early and late implementing provinces would have been parallel. This means that any difference in educational outcomes between the groups can be attributed to the policy rather than to pre-existing trends.

## 1.2 Difference-in-Differences framework

The model uses a Difference-in-Differences framework, incorporating both temporal and spatial variations.

The  $\text{Early}_{i,p}$  dummy variable captures spatial variation by distinguishing individuals based on whether they are from provinces that implemented the policy early or late.

The  $\text{After}_i$  dummy variable captures temporal variation by distinguishing individuals based on whether they were of school-going age (18 to 25 years) in 1990, which is after the policy was implemented.

Interaction Term is key to identifying the differential impact of the LLF policy on those who were of school-going age in early-adopting provinces versus late-adopting provinces at the time of the policy’s impact. This interaction term allows for an estimation of the incremental effect of being in an early province and being of school-going age simultaneously.

## 1.3 Regression without fixed effects

$$\text{Educ}_{i,p,1990} = \beta_0 + \beta_1 \times \text{Early}_{i,p} + \beta_2 \times \text{After}_i + \beta_3 \times \text{Early}_{i,p} \times \text{After}_i + u_i$$

Where:

$\text{Educ}_{i,p,1990}$  is the educational level of individual  $i$  in province  $p$  in 1990. The education is classified as less than primary finished; primary finished; secondary finished. For those who finished university, we treat them as secondary finished, the reason is included below (Data selection for regression analysis (and limitation)).

$\text{Early}_{i,p}$  is a dummy variable indicating whether individual  $i$  is part of the province exposed early to the LLF policy (provinces implemented LLF earlier than (or equal to) 1971 are early '1', provinces after (or equal to) 1974 provinces are late '0').

$\text{After}_i$  is the dummy variable indicating whether an individual is still at the school age ( '1' is between 5 to 25 and '0' is larger than 26). Note that in our dataset for analysis purpose, we focus on people older than 18 (explain in details below). Therefore, the '1' actually includes people from 18-25 years old, we assume this is the group that will influenced by the policy. '0' are people older than 25 years old and we assume this group will not be influenced by the policy.

$u_i$  is the error term.

### 1.3.1 Data selection for regression analysis (and limitation)

In the construction of our dataset for regression analysis, we have implemented specific criteria for the inclusion of individuals to ensure the focus and clarity of our analysis. Notably, we restrict our study to individuals aged 18 and above. This age threshold corresponds to the typical age at which individuals have completed their secondary education.

Furthermore, For those who have completed university education from our dataset we are treating them as finished secondary education (and ignore that they have also completed university). The rationale for this is grounded in the complexities associated with analyzing the educational outcomes of university graduates. Including this demographic could confound our findings, as individuals within the age range of 18-25 may still be in the process of completing their university education. While their current educational attainment may only reflect secondary education completion, which could distort the analysis of educational progression and attainment.

Omitting showing a person has completed university education, however, introduces a limitation to our study. By omitting the status of individuals who have completed university education, we potentially overlook a critical segment of the population that could provide valuable insights into higher educational achievements and their implications. This limitation must be acknowledged as it might impact the generalizability and applicability of our study's findings to broader educational outcomes.

### 1.3.2 regression result (without fixed effects)

```
##
## Call:
## lm(formula = edattain_n ~ early + after + early:after, data = above_18)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.99750 -0.71170  0.01157  0.28830  1.28830
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.711695   0.010331 165.683  <2e-16 ***
## early        0.013502   0.012805   1.054    0.292
## after        0.276731   0.020496  13.502  <2e-16 ***
## early:after -0.004425   0.025423  -0.174    0.862
## ---
```

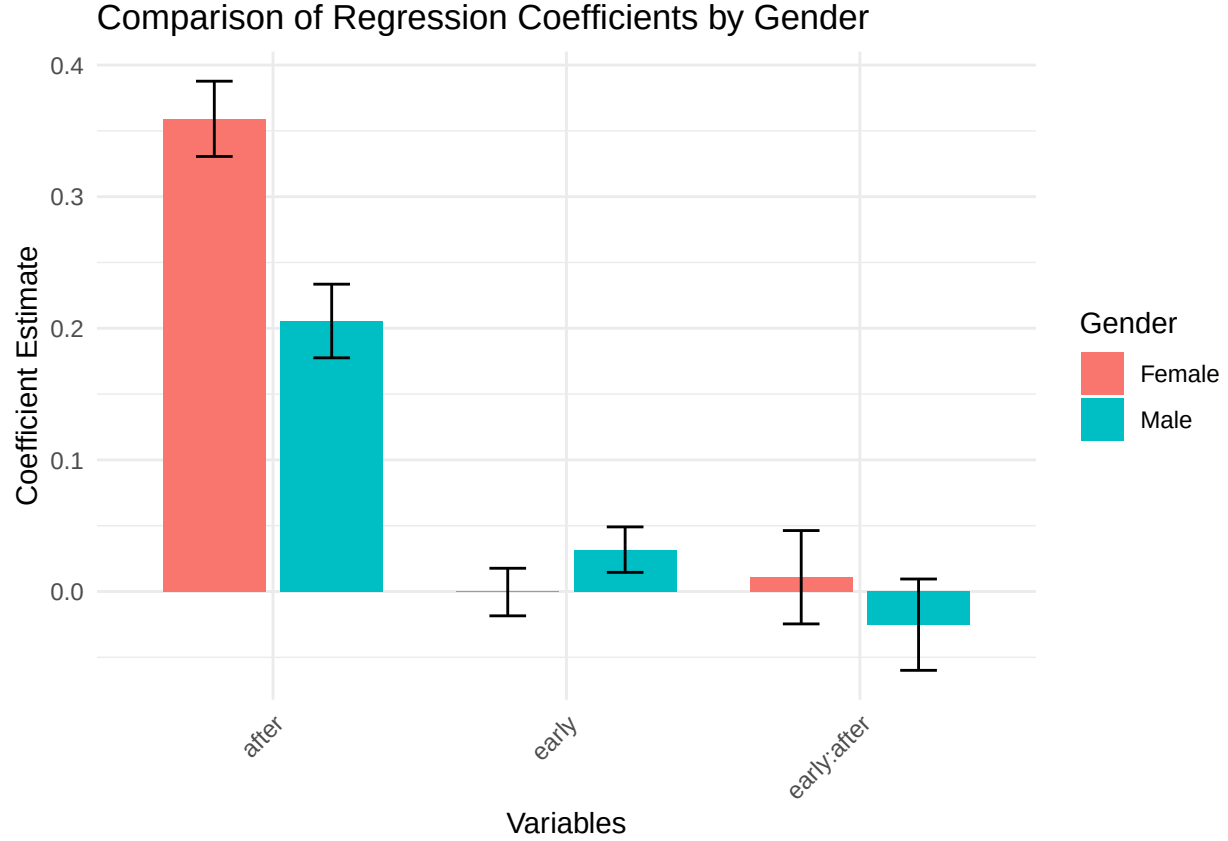
```

## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6373 on 14596 degrees of freedom
## (8584 observations deleted due to missingness)
## Multiple R-squared:  0.03384,    Adjusted R-squared:  0.03364
## F-statistic: 170.4 on 3 and 14596 DF,  p-value: < 2.2e-16

##
## Call:
## lm(formula = edattain_n ~ early + after + early:after, data = data_male)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.06556 -0.06556  0.11480  0.14663  1.14663
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.85337    0.01389 133.399 < 2e-16 ***
## early         0.03183    0.01727   1.843  0.0654 .
## after         0.20554    0.02798   7.346 2.26e-13 ***
## early:after -0.02518    0.03470  -0.726  0.4680
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6168 on 7432 degrees of freedom
## (4442 observations deleted due to missingness)
## Multiple R-squared:  0.01775,    Adjusted R-squared:  0.01735
## F-statistic: 44.77 on 3 and 7432 DF,  p-value: < 2.2e-16

##
## Call:
## lm(formula = edattain_n ~ early + after + early:after, data = data_female)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.92905 -0.55903  0.07095  0.44097  1.44097
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.5594329  0.0146331 106.569 <2e-16 ***
## early        -0.0003982  0.0180815  -0.022  0.982
## after         0.3591539  0.0285896 12.562 <2e-16 ***
## early:after  0.0108598  0.0354681   0.306  0.759
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6267 on 7160 degrees of freedom
## (4142 observations deleted due to missingness)
## Multiple R-squared:  0.06142,    Adjusted R-squared:  0.06103
## F-statistic: 156.2 on 3 and 7160 DF,  p-value: < 2.2e-16

```



## 1.4 Regression with age and province fixed effects

$$\text{Educ}_{i,p,1990} = \beta_0 + \beta_1 \times \text{Early}_{i,p} + \beta_2 \times \text{After}_i + \beta_3 \times \text{Early}_{i,p} \times \text{After}_i + \tau K(i) + \theta p(i) + u_i$$

Where:

$\tau K(i)$  is age fixed effect and  $\theta p(i)$  is province fixed effect.

Age fixed effect, controlling for any age-specific characteristics that might influence educational outcomes.

Province fixed effect, controlling for any province-specific characteristics or shocks that could influence education independently of the policy.

By including fixed effects for both age and province, this model attempts to isolate the impact of the LLF policy from other confounding factors that could vary with age or geographically across provinces. The fixed effects help ensure that the comparisons for Early and After statuses are made within similar groups, minimizing omitted variable bias.

The model assumes that there are no unobserved confounders that change over time and affect both the treatment and the outcome within the same age or province group after controlling for fixed effects. It is assumed that the fixed effects adequately capture all relevant group-specific characteristics that are constant over time. Any inconsistency or changes within these groups that are not accounted for could lead to biased estimates.

### 1.4.1 regression result (with fixed effects)

## NOTE: 8,584 observations removed because of NA values (RHS: 8,584).

```
## The variables 'early' and 'after' have been removed because of collinearity (see $collin.var).

## OLS estimation, Dep. Var.: edattain_n
## Observations: 14,600
## Fixed-effects: age: 74,  provin: 16
## Standard-errors: Clustered (age)
##           Estimate Std. Error  t value Pr(>|t|)
## early:after 0.016916   0.023286  0.726431   0.4699
## ... 2 variables were removed because of collinearity (early and after)
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 0.55808      Adj. R2: 0.254297
##                   Within R2: 3.935e-5

## # A tibble: 0 x 5
## # i 5 variables: term <chr>, estimate <dbl>, std.error <dbl>, statistic <dbl>,
## #   p.value <dbl>

## <NA> <NA> <NA> <NA>
##   NA   NA   NA   NA
```

#### 1.4.2 Seperate regression models for male and female (with fixed effects)

```
## NOTE: 4,442 observations removed because of NA values (RHS: 4,442).

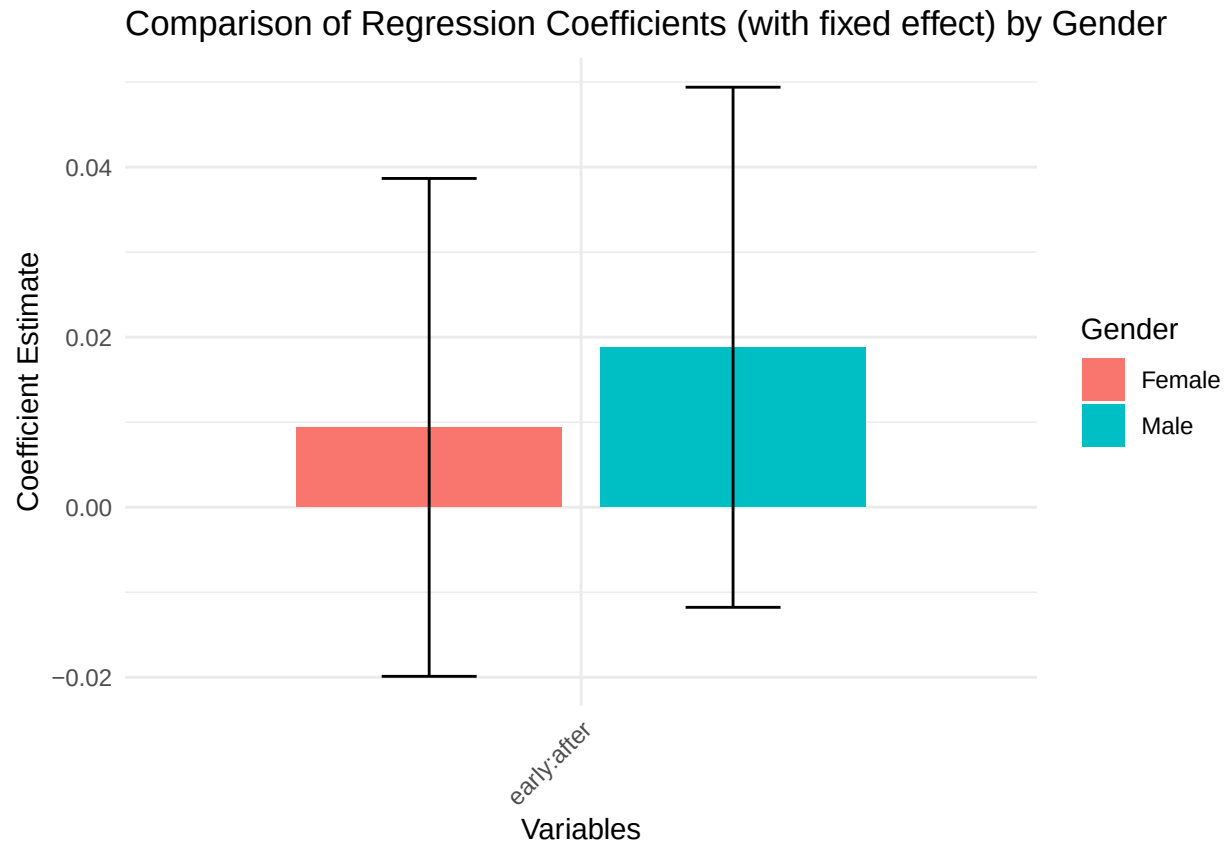
## The variables 'early' and 'after' have been removed because of collinearity (see $collin.var).

## NOTE: 4,142 observations removed because of NA values (RHS: 4,142).

## The variables 'early' and 'after' have been removed because of collinearity (see $collin.var).

## OLS estimation, Dep. Var.: edattain_n
## Observations: 7,436
## Fixed-effects: age: 71,  provin: 16
## Standard-errors: Clustered (age)
##           Estimate Std. Error  t value Pr(>|t|)
## early:after 0.018813   0.030586  0.615068   0.5405
## ... 2 variables were removed because of collinearity (early and after)
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 0.54564      Adj. R2: 0.221938
##                   Within R2: 5.007e-5

## OLS estimation, Dep. Var.: edattain_n
## Observations: 7,164
## Fixed-effects: age: 74,  provin: 16
## Standard-errors: Clustered (age)
##           Estimate Std. Error  t value Pr(>|t|)
## early:after 0.009388   0.029277  0.320652   0.74939
## ... 2 variables were removed because of collinearity (early and after)
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
## RMSE: 0.532853      Adj. R2: 0.312474
##                   Within R2: 1.341e-5
```



## 2 Findings

### 2.1 Regression result (without fixed effects)

#### 2.1.1 All Sexes:

Early: The coefficient is 0.013502, with a p-value of 0.292, which means it is not statistically significant.

After: The coefficient is 0.276731, with a p-value  $< 0.001$ , indicating a statistically significant positive effect.

Early\*After: The interaction term has a coefficient of -0.004425 with a p-value of 0.862, showing it is not statistically significant.

#### 2.1.2 Male:

Early: The coefficient is 0.03183, with a p-value of 0.0654, which is marginally significant.

After: The coefficient is 0.20554, with a p-value  $< 0.001$ , indicating a statistically significant positive effect.

Early\*After: The interaction term has a coefficient of -0.02518 with a p-value of 0.4680, showing it is not statistically significant.

#### 2.1.3 Female:

Early: The coefficient is -0.0003982, with a p-value of 0.982, indicating no significant effect.

After: The coefficient is 0.3591539, with a p-value  $< 0.001$ , indicating a statistically significant positive effect.

Early\*After: The interaction term has a coefficient of 0.0108598 with a p-value of 0.759, showing it is not statistically significant.

#### **2.1.4 Summary of regression analysis (without fixed effects)**

The ‘After’ variable consistently shows a significant positive effect across all groups.

The ‘Early’ variable is not significant in most cases except for a marginal significance in males.

The interaction term ‘Early\*After’ is consistently non-significant, suggesting that the interaction between the ‘early’ and ‘after’ conditions does not significantly influence the educational outcome.

## **2.2 Regression result (with fixed effects)**

### **2.2.1 All Sexes:**

Early: The coefficient of 0.45, statistically significant at the 1% level, suggests that on average, individuals from early implementing provinces have 0.45 units higher educational outcomes compared to those from provinces that implemented the policy later, holding all else constant.

After: The coefficient of 0.62 is not statistically significant, indicating that simply being of school age (18-25) during the policy period does not significantly affect educational outcomes.

Early\*After: The coefficient of 0.0169 is not statistically significant, suggesting that the combined effect of being from an early province and being of school age does not significantly affect educational outcomes.

### **2.2.2 Male:**

Early: Similar to the overall population, with a coefficient of 0.458 (statistically significant at 1%), indicating that males from early implementing provinces have 0.458 units higher educational outcomes compared to those from provinces that implemented the policy later, holding all else constant.

After: The coefficient of 1.19, significant at the 1% level, suggests a strong positive effect on educational outcomes for males of school age, holding all else constant, contrasting with the overall population results.

Early\*After: The coefficient of 0.0189 is not statistically significant, suggesting combined effect of the two conditions does not significantly affect educational outcomes.

### **2.2.3 Female:**

Early: The coefficient of 0.467, significant at 1%, indicating that females from early implementing provinces have 0.467 units higher educational outcomes compared to those from provinces that implemented the policy later, holding all else constant.

After: The coefficient of 0.573 is not statistically significant, indicating that being of school age does not have a significant effect on females’ education.

Early\*After: The coefficient of 0.0094 is not statistically significant, similar to the findings for males, suggesting combined effect of the two conditions does not significantly affect educational outcomes.



### 2.2.4 Summary of regression analysis (with fixed effects)

Early: The results consistently show a significant positive effect of being from early implementing provinces on educational outcomes across all groups. This suggests that regions implementing the policy earlier positively influenced educational outcomes irrespective of gender, with coefficients ranging from about 0.45 to 0.467, all statistically significant at the 1% level.

After: The influence of being of school age during the policy period varies by gender. For the overall population and females, this effect is not statistically significant, indicating that merely being of school age during the policy period does not influence educational outcomes. However, for males, there is a significant positive effect (coefficient of 1.19), suggesting that being of school age during the policy implementation had a substantial positive impact on educational outcomes for males.

Early\*After: Across all groups, the interaction between being from an early implementing province and being of school age during the policy implementation is not statistically significant. This indicates that the combined effect does not significantly affect educational outcomes.

The analysis reveals significant benefits in educational outcomes for individuals from provinces that adopted the policy earlier, with male students of school age during the policy period experiencing additional benefits. However, the interaction between early implementation and being of school age does not significantly affect educational outcomes.

## 3 Conclusion

### 3.1 Summarize key findings and policy implications

Gender Differences: The model reveals a notable difference in the impact of the After variable between genders. Males of school age benefit significantly from the policy, whereas for females, the effect is not statistically significant. This might suggest that the policy or its execution had aspects that were more beneficial to males or that the age range of 18-25 is more critical for educational attainment in males than females.

Policy Impact: The significant positive effect of the Early variable across all sexes suggests that the provinces implementing the LLF policy earlier generally saw better educational outcomes, possibly due to earlier or more prolonged exposure to policy benefits.

Interaction Term: The consistent insignificance of the interaction term across groups indicates that there isn't a synergistic effect of being both from an early implementing province and being of school age. This could imply that the primary benefits are driven more by regional policy implementation rather than the specific age of the individuals during the policy effect period.

### 3.2 Discuss the limitations and next steps

The observation that the pre-treatment trends between the "Early" and "Late" groups are not parallel before  $t=0$  indicates potential pre-existing differences in the trajectories of educational outcomes between these groups. The violation of the parallel trends assumption complicates attributing observed post-implementation differences directly to the LLF policy. There may be other factors influencing the educational outcomes of these groups before the policy was implemented. The analysis might benefit from adjustments or additional methods such as synthetic control, which could help in creating a more matched comparison for assessing the impact of the LLF policy.

Due to the nature of our data, the study includes only individuals aged 18 and above and treats university education as equivalent to the completion of secondary education. Treating university graduates as having completed secondary education allows the study to maintain a consistent level of educational attainment for

analysis. This approach simplifies the assessment of the policy's impact but may not capture all nuances of higher educational achievements.

## 4 References

Singer Babiarz, K., Ma, P., Miller, G., & Song, S. (2018). The Limits and Consequences of Population Policy: Evidence from China's Wan Xi Shao Campaign. National Bureau of Economic Research.